

Qns	Answer	Marks
6(a)	Faraday's Law: Induced emf is directly proportional to the rate of change of magnetic flux linkage	B1
	Lenz's Law: Direction of induced emf produces effects to oppose the change causing it	B1
6(b)	when pen is near to the screen, the <u>changing magnetic field causes a changing magnetic flux linkage</u> through the wire loops inside the pen	B1
	using Faraday's law, <u>e.m.f. is induced in the loop, which results in a current due to a closed circuit</u> to operate the pen.	B1
6(c)(i)	By Faraday's Law: induced e.m.f. = $\frac{d}{dt} NBA$ magnitude of peak e.m.f. induced = $NA \left \frac{dB}{dt} \right _{\max}$ $= NA \left 2\pi f B_0 \cos(2\pi ft) \right _{\max}$ $= NA (2\pi f B_0)$ $= 100(3 \times 10^{-6}) [2\pi (750 \times 10^3) B_0]$ $100(3 \times 10^{-6}) [2\pi (750 \times 10^3) B_0] = 0.80 \text{ V}$ $B_0 = 5.658 \times 10^{-4} \text{ T}$ $= 5.7 \times 10^{-4} \text{ T}$	C1 A1
6(c)(ii)	<u>the rate of change of magnetic flux linkage with Earth's magnetic flux density due to movement of hand is much lower than that from the screen due to the high frequency of 750 khz</u> so induced e.m.f. due to Earth's magnetic flux density insignificant relative to that due to screen.	B1
6(c)(iii)	average power = $\frac{V_{rms}^2}{R}$ $= \frac{\left(\frac{V_{\max}}{\sqrt{2}} \right)^2}{2.0}$ $= \frac{\left(\frac{0.80}{\sqrt{2}} \right)^2}{2.0}$ $= 0.16 \text{ W}$	C1 A1

Qns	Answer	Marks
7(a)(i)		